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Probabilistic Topic Models

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- Introduction
 - Recap: Probability distributions for text
- Topic Models
 - Inference
 - Model Evaluation
 - Model Practicalities
- Constraining topic models
- Structural Topic Models



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After this lecture, you should be able to

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- Explain the intuition behind LDA as a model for recurring word co-occurrence



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- Interpret Θ as document-topic proportions and Φ as topic-word probabilities



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- Make and document key corpus, document-unit, and preprocessing choices



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 - Evaluate a topic model using words, documents, stability checks, and research purpose



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- Interpret Θ as document-topic proportions and Φ as topic-word probabilities
- Make and document key corpus, document-unit, and preprocessing choices
- Evaluate a topic model using words, documents, stability checks, and research purpose
- Decide when a topic model is useful, and when it is the wrong tool



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Section 1

Introduction



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Why topic models?

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- Study semantic themes in a corpus (or topics)



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- Study semantic themes in a corpus (or topics)
- Exploratory (unsupervised) analysis



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- Study semantic themes in a corpus (or topics)
- Exploratory (unsupervised) analysis
- (Relatively) simple statistical/probabilistic models



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- Exploratory (unsupervised) analysis
- (Relatively) simple statistical/probabilistic models
- Transparent models with statistical guarantees



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- Study semantic themes in a corpus (or topics)
- Exploratory (unsupervised) analysis
- (Relatively) simple statistical/probabilistic models
- Transparent models with statistical guarantees
- Has been extended in a large number of ways



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Topic models as measurement

Topic models are **not** topic discovery machines.

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Topic models are **not** topic discovery machines.

They are measurement **models** for recurring patterns of word co-occurrence.

- The model gives us a structured representation of a corpus

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- The model gives us a structured representation of a corpus
- The labeling and interpretation comes from the researcher



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- The model gives us a structured representation of a corpus
- The labeling and interpretation comes from the researcher
- The result is useful only if it is validated for the research question



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Topic modeling workflow

1. Research question

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Topic modeling workflow

1. Research question
2. Choose corpus

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Topic modeling workflow

1. Research question
2. Choose corpus
3. Decide document unit

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Topic modeling workflow

1. Research question
2. Choose corpus
3. Decide document unit
4. Pre-process textual content

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Topic modeling workflow

1. Research question
2. Choose corpus
3. Decide document unit
4. Pre-process textual content
5. Model settings

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Topic modeling workflow

1. Research question
2. Choose corpus
3. Decide document unit
4. Pre-process textual content
5. Model settings
6. Estimate model

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1. Research question
2. Choose corpus
3. Decide document unit
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7. Validate/critisise/interpret (if problem - go to 5, or further back)

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8. Use model



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1. Research question
 2. Choose corpus
 3. Decide document unit
 4. Pre-process textual content
 5. Model settings
 6. Estimate model
 7. Validate/critisise/interpret (if problem - go to 5, or further back)
 8. Use model
- Every step is a research-design decision



Probability models for text: Multinomial

- Let θ be the probability of drawing the **word type** k . e.g.

$$\theta = (\text{monkey} = 0.001, \text{the} = 0.03, \dots, \text{Norrköping} = 0.0001)$$

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- We are interested in a probability model for word (tokens), namely

$$w \sim p(w|\theta)$$

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- The Multinomial/Categorical distribution, where

$$p(w|\theta) = \theta_1^{w_1} \cdot \dots \cdot \theta_K^{w_K}$$

where $\sum_{k=1}^K \theta_k = 1$

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- A probabilistic **unigram** language model



Probability models for text: Dirichlet

- We are interested in a probability model for word probabilities (θ), namely

$$\theta \sim p(\theta|\alpha)$$

- The Dirichlet, where

$$p(\theta|\alpha) = \frac{\Gamma(\sum_{k=1}^K \alpha_k)}{\prod_{k=1}^K \Gamma(\alpha_k)} \prod_{k=1}^K \theta_k^{\alpha_k - 1},$$

and $\sum_{k=1}^K \theta_k = 1$, $\alpha > 0$ and Γ is the gamma function.

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- The Dirichlet distribution generates "probability distributions", e.g.

$$\theta_1 = (0.019, 0.021, \dots, 0.0002)$$

$$\theta_2 = (0.012, 0.019, \dots, 0.0001)$$

...

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Estimating parameters

- Maximum likelihood for word type v is estimated as

$$\hat{\theta}_{v,MLE} = \frac{n_v}{\sum_v n_v},$$

where n_v are the sufficient statistics (or word counts).

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- The Dirichlet is a conjugate prior for the Multinomial distribution, so:
- Using the Multinomial (likelihood) and Dirichlet (prior), we get

$$\theta \sim p(\theta|w, \alpha) = \text{Dir}(\alpha + \mathbf{n}_v),$$

where θ is a vector of length V and the prior hyperparameter α can be seen as a smoothing constant.



Example

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$\mathbf{w}_1$	boat	shore	bank		
$\mathbf{w}_2$	Zlatan	boat	shore	money	bank
$\mathbf{w}_3$	money	bank	soccer	money	

MLE:

$$\hat{\theta}_{\text{bank}, MLE} = \frac{n_v}{\sum_v n_v} = \frac{3}{12} = 0.25,$$

Posterior:

$$p(\theta_{\text{boat}, \dots, \text{bank}} | \mathbf{w}, \alpha) = \text{Dir}(\alpha + (2, \dots, 3)),$$



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Section 2

Topic Models



The Latent Dirichlet Allocation model

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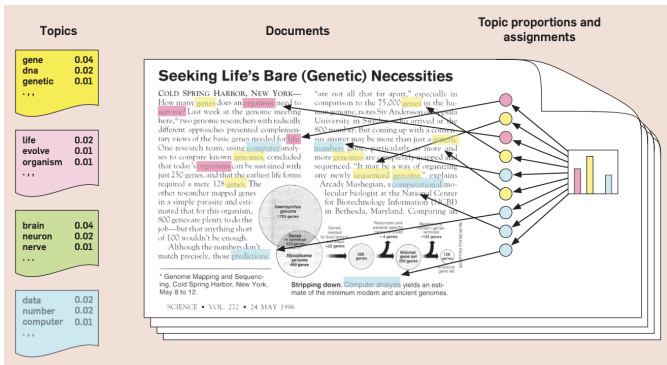


Figure: The intuitions behind latent Dirichlet allocation (Blei, 2012)



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Latent Dirichlet Allocation

1. For each k in $1 \dots K$:
 - 1.1 $\phi_k \sim \text{Dirichlet}(\beta)$
2. For each document d in $1 \dots D$:
 - 2.1 $\theta_d \sim \text{Dirichlet}(\alpha)$
 - 2.2 For each word i :
 - 2.2.1 $z_{id} \sim \text{Categorical}(\theta_d)$
 - 2.2.2 $w_{id} \sim \text{Categorical}(\phi_{z_{id}})$

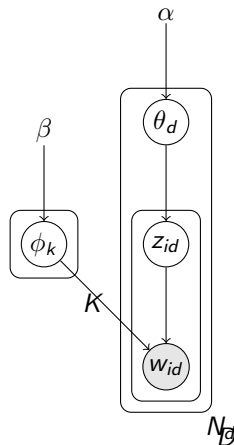


Figure: Probabilistic model for Latent Dirichlet Allocation (LDA)



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Corpus and document design

- A topic model estimates **document** patterns within the corpus you give it

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Corpus and document design

- A topic model estimates **document** patterns within the corpus you give it
- The document unit changes the measurement target
 - Book, chapter, article, paragraph, speech, interview answer, social media post
- **What to choose and why?**

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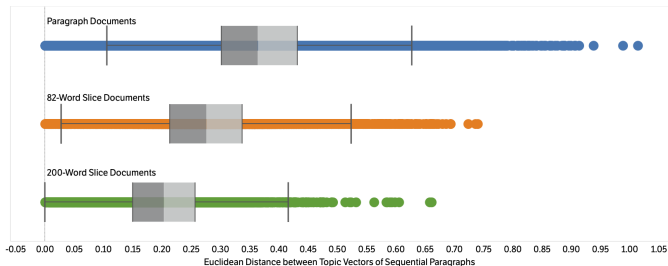


Figure: Algee-Hewitt et al (2015), Fig. 6.1



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Preprocessing as measurement

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- Preprocessing changes what the model can see



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- Preprocessing changes what the model can see
 - Decide and document:
 - Tokenization, stop words, rare-word thresholds



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- **Ask:** does this choice remove noise, or remove part of the phenomenon?



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 - **Ask:** does this choice remove noise, or remove part of the phenomenon?
 - lemmatize: e.g. 'running' to 'run' ?
 - remove names?



Example of parameters $\mathbf{z}$, Θ and Φ

$\mathbf{w}_1$	boat	shore	bank		
$\mathbf{z}_1$	1	1	1		
$\mathbf{w}_2$	Zlatan	boat	shore	money	bank
$\mathbf{z}_2$	2	1	1	3	3
$\mathbf{w}_3$	money	bank	soccer	money	
$\mathbf{z}_3$	3	3	2	3	

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		boat	shore	soccer	Zlatan	bank	money
$\Phi =$	Topic 1	0.35	0.35	0.05	0.05	0.15	0.05
	Topic 2	0.025	0.025	0.45	0.45	0.025	0.025
	Topic 3	0.025	0.025	0.025	0.025	0.45	0.45



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	Topic 3	0.025	0.025	0.025	0.025	0.45	0.45

		Topic 1	Topic 2	Topic 3
$\Theta =$	doc 1	0.96	0.02	0.02
	doc 2	0.3	0.2	0.5
	doc 3	0.05	0.35	0.6



A small topic model example

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Closing arguments were heard yesterday in the Federal bankruptcy fraud trial of Stephen J. Sabbeth, whose legal problems have raised doubts about his ability to continue as leader of the Nassau County Democratic Party.

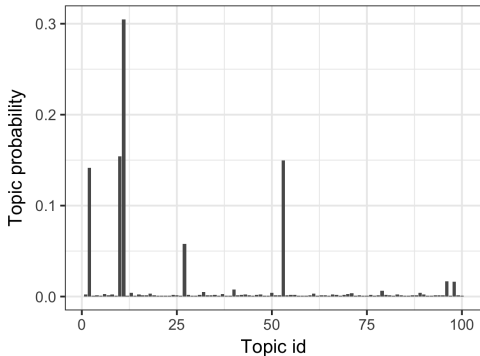
Mr. Sabbeth is charged with trying to conceal \$750,000 from his bank creditors by hiding the money in a secret account in his wife's maiden name, rather than use it to pay creditors when his lumber business went into bankruptcy 10 years ago.

– The New York Times 25th of February 1999



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Estimated topic distribution $E(\theta_d)$



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Most probable word type by topic

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Topic	Top words (by ϕ_{kv})
2	party election voters campaign democratic
10	bank banks loans loan insurance savings
11	trial prison jury prosecutors convicted guilty
53	investigation inquiry documents investigators

Table: Most probable words in topic 2, 10, 11 and 53.



Analytical use of topic models

- We have three parameters we can use for inference, Φ , Θ , and $\mathbf{z}$

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Analytical use of topic models

- We have three parameters we can use for inference, Φ , Θ , and $\mathbf{z}$

		boat	shore	soccer	Zlatan	bank	money
$\Phi =$	Topic 1	0.35	0.35	0.05	0.05	0.15	0.05
	Topic 2	0.025	0.025	0.45	0.45	0.025	0.025
	Topic 3	0.025	0.025	0.025	0.025	0.45	0.45

		Topic 1	Topic 2	Topic 3
$\Theta =$	doc 1	0.96	0.02	0.02
	doc 2	0.3	0.2	0.5
	doc 3	0.05	0.35	0.6

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Analytical use of topic models

- We have three parameters we can use for inference, Φ , Θ , and $\mathbf{z}$

$\Phi =$

	boat	shore	soccer	Zlatan	bank	money
Topic 1	0.35	0.35	0.05	0.05	0.15	0.05
Topic 2	0.025	0.025	0.45	0.45	0.025	0.025
Topic 3	0.025	0.025	0.025	0.025	0.45	0.45

$\Theta =$

	Topic 1	Topic 2	Topic 3
doc 1	0.96	0.02	0.02
doc 2	0.3	0.2	0.5
doc 3	0.05	0.35	0.6

$\mathbf{z}_1$	1	1	1		
$\mathbf{w}_1$	boat	shore	bank		
$\mathbf{z}_2$	2	1	1	3	3
$\mathbf{w}_2$	Zlatan	boat	shore	money	bank
$\mathbf{z}_3$	3	3	2	3	
$\mathbf{w}_3$	money	bank	soccer	money	



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Subsection 1

Inference



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What to remember

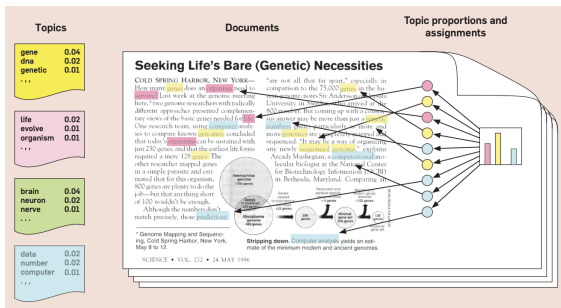


Figure: The intuitions behind latent Dirichlet allocation (Blei, 2012)

- LDA assumes each document mixes multiple topics



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What to remember

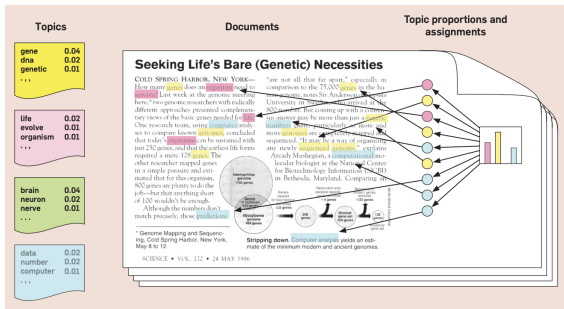


Figure: The intuitions behind latent Dirichlet allocation (Blei, 2012)

- LDA assumes each document mixes multiple topics
- Each topic is a probability distribution over words



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What to remember

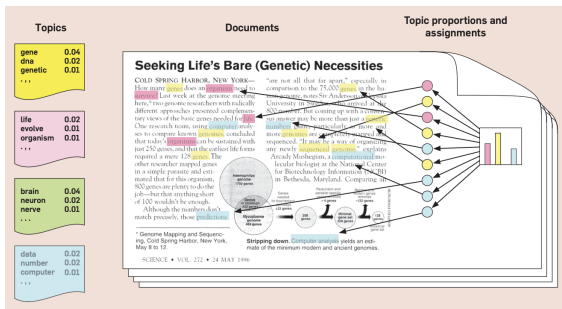


Figure: The intuitions behind latent Dirichlet allocation (Blei, 2012)

- LDA assumes each document mixes multiple topics
- Each topic is a probability distribution over words
- Inference estimates the hidden topic assignments that make the observed words plausible



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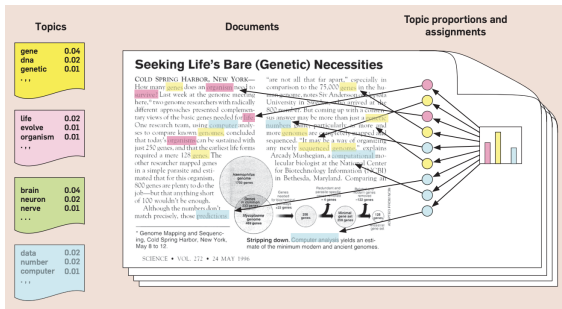


Figure: The intuitions behind latent Dirichlet allocation (Blei, 2012)

- LDA assumes each document mixes multiple topics
- Each topic is a probability distribution over words
- Inference estimates the hidden topic assignments that make the observed words plausible
- However, the key question is whether the estimated structure **measures something meaningful**



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Inference methods

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- We want to estimate the parameters $\mathbf{z}, \theta, \phi$ based on data $\mathbf{w}$



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- We want to estimate the parameters $\mathbf{z}, \theta, \phi$ based on data $\mathbf{w}$
- Most commonly done using Bayesian methods by computing the joint posterior

$$p(\mathbf{z}, \theta, \phi | \mathbf{w}) = \frac{p(\mathbf{w} | \mathbf{z}, \theta, \phi) p(\mathbf{z}, \theta, \phi)}{p(\mathbf{w})}$$



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- We want to estimate the parameters $\mathbf{z}, \theta, \phi$ based on data $\mathbf{w}$
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$$p(\mathbf{z}, \theta, \phi | \mathbf{w}) = \frac{p(\mathbf{w} | \mathbf{z}, \theta, \phi) p(\mathbf{z}, \theta, \phi)}{p(\mathbf{w})}$$

- $p(\mathbf{w})$ is intractable, so we use
 - MCMC/Gibbs sampling (Griffiths and Steyvers, 2004), or
 - Variational inference (Blei, Ng and Jordan, 2003).



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Sufficient statistics

$\mathbf{w}_1$	boat	shore	bank		
$\mathbf{z}_1$	1	1	1		
$\mathbf{w}_2$	Zlatan	boat	shore	money	bank
$\mathbf{z}_2$	1	1	1	3	3
$\mathbf{w}_3$	money	bank	soccer	money	
$\mathbf{z}_3$	3	3	2	3	

		boat	shore	soccer	Zlatan	bank	money
$\mathbf{n}_v =$	Topic 1	2	2	0	1	1	0
	Topic 2	0	0	1	0	0	0
	Topic 3	0	0	0	0	2	3



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$\mathbf{w}_1$	boat	shore	bank			
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$\mathbf{z}_2$	1	1	1	3	3	
$\mathbf{w}_3$	money	bank	soccer	money		
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		boat	shore	soccer	Zlatan	bank	money
$\mathbf{n}_v =$	Topic 1	2	2	0	1	1	0
	Topic 2	0	0	1	0	0	0
	Topic 3	0	0	0	0	2	3

		Topic 1	Topic 2	Topic 3
$\mathbf{n}_d =$	doc 1	3	0	0
	doc 2	3	0	2
	doc 3	0	1	3



- Gibbs sampling (Griffiths and Steyvers, 2004)

$$p(z_i = k) \propto \theta_{d(i),k} \phi_{k,v(i)}$$

$$\theta_d \sim \text{Dir}(\mathbf{n}_d + \alpha)$$

$$\phi_k \sim \text{Dir}(\mathbf{n}_v + \beta)$$

where n_d and n_v are **sufficient** statistics for θ and ϕ

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- Gibbs sampling (Griffiths and Steyvers, 2004)

$$p(z_i = k) \propto \theta_{d(i),k} \phi_{k,v(i)}$$

$$\theta_d \sim \text{Dir}(\mathbf{n}_d + \alpha)$$

$$\phi_k \sim \text{Dir}(\mathbf{n}_v + \beta)$$

where n_d and n_v are **sufficient** statistics for θ and ϕ

- Run until convergence (commonly log-likelihood convergence)
- Then the MCMC/Gibbs sampler generates draws from

$$p(\mathbf{z}, \theta, \phi | \mathbf{w})$$



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- We can integrate out Θ and Φ

$$\begin{aligned} p(z_i = k | \mathbf{z}_{-i}) &\propto \frac{n_{dk} + \alpha}{\sum^K (n_{dk} + \alpha)} \frac{n_{vk} + \beta}{\sum^V (n_{vk} + \beta)} \\ &\propto (n_{dk} + \alpha) \frac{n_{vk} + \beta}{\sum^V (n_{vk} + \beta)} \end{aligned}$$



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- We can integrate out Θ and Φ

$$\begin{aligned} p(z_i = k | \mathbf{z}_{-i}) &\propto \frac{n_{dk} + \alpha}{\sum^K (n_{dk} + \alpha)} \frac{n_{vk} + \beta}{\sum^V (n_{vk} + \beta)} \\ &\propto (n_{dk} + \alpha) \frac{n_{vk} + \beta}{\sum^V (n_{vk} + \beta)} \end{aligned}$$

- This is the **collapsed Gibbs sampler** for LDA



Collapsed Gibbs sampling

Let $\alpha = \beta = 0.5$, and we will sample $\mathbf{z}_2$:

$$p(z_i = k) \propto \left(\frac{2.5}{5.5} \frac{0.5}{1.5}, \frac{0.5}{5.5} \frac{0.5}{1.5}, \frac{2.5}{5.5} \frac{0.5}{1.5} \right)$$

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$\mathbf{w}_1$	boat	shore	bank		
$\mathbf{z}_1$	1	1	1		
$\mathbf{w}_2$	Zlatan	boat	shore	money	bank
$\mathbf{z}_2$	?	1	1	3	3
$\mathbf{w}_3$	money	bank	soccer	money	
$\mathbf{z}_3$	3	3	2	3	

		boat	shore	soccer	Zlatan	bank	money
$\mathbf{n}_v =$	Topic 1	2	2	0	0	1	0
	Topic 2	0	0	1	0	0	0
	Topic 3	0	0	0	0	2	3

		Topic 1	Topic 2	Topic 3
$\mathbf{n}_2 =$	doc 2	2	0	2



Collapsed Gibbs sampling

Let $\alpha = \beta = 0.5$, and we will sample $\mathbf{z}_2$:

$$p(z_i = k) \propto \left(\frac{1.5}{5.5} \frac{1.5}{2.5}, \frac{0.5}{5.5} \frac{0.5}{2.5}, \frac{3.5}{5.5} \frac{0.5}{2.5} \right)$$

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$\mathbf{w}_1$	boat	shore	bank		
$\mathbf{z}_1$	1	1	1		
$\mathbf{w}_2$	Zlatan	boat	shore	money	bank
$\mathbf{z}_2$	3	?	1	3	3
$\mathbf{w}_3$	money	bank	soccer	money	
$\mathbf{z}_3$	3	3	2	3	

		boat	shore	soccer	Zlatan	bank	money
$\mathbf{n}_v =$	Topic 1	1	2	0	0	1	0
	Topic 2	0	0	1	0	0	0
	Topic 3	0	0	0	1	2	3

		Topic 1	Topic 2	Topic 3
$\mathbf{n}_2 =$	doc 2	1	0	3



Collapsed Gibbs sampling

Let $\alpha = \beta = 0.5$, and we will sample $\mathbf{z}_2$:

$$p(z_i = k) \propto \left(\frac{1.5}{5.5} \frac{1.5}{2.5}, \frac{0.5}{5.5} \frac{0.5}{2.5}, \frac{3.5}{5.5} \frac{0.5}{2.5} \right)$$

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	Topic 3	0	0	0	1	2	3

		Topic 1	Topic 2	Topic 3
$\mathbf{n}_2 =$	doc 2	1	0	3



Collapsed Gibbs sampling

Let $\alpha = \beta = 0.5$, and we will sample $\mathbf{z}_2$:

$$p(z_i = k) \propto \left(\frac{1.5}{5.5} \frac{0.5}{3.5}, \frac{0.5}{5.5} \frac{0.5}{3.5}, \frac{3.5}{5.5} \frac{2.5}{3.5} \right)$$

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$\mathbf{w}_1$	boat	shore	bank		
$\mathbf{z}_1$	1	1	1		
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$\mathbf{z}_2$	3	1	3	?	3
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$\mathbf{z}_3$	3	3	2	3	

		boat	shore	soccer	Zlatan	bank	money
$\mathbf{n}_v =$	Topic 1	2	1	0	0	1	0
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	Topic 3	0	1	0	1	2	2

		Topic 1	Topic 2	Topic 3
$\mathbf{n}_2 =$	doc 2	1	0	3



Collapsed Gibbs sampling

Let $\alpha = \beta = 0.5$, and we will sample $\mathbf{z}_2$:

$$p(z_i = k) \propto \left(\frac{1.5}{5.5} \frac{1.5}{3.5}, \frac{0.5}{5.5} \frac{0.5}{3.5}, \frac{3.5}{5.5} \frac{1.5}{3.5} \right)$$

$\mathbf{w}_1$	boat	shore	bank		
$\mathbf{z}_1$	1	1	1		
$\mathbf{w}_2$	Zlatan	boat	shore	money	bank
$\mathbf{z}_2$	3	1	3	3	?
$\mathbf{w}_3$	money	bank	soccer	money	
$\mathbf{z}_3$	3	3	2	3	

		boat	shore	soccer	Zlatan	bank	money
$\mathbf{n}_v =$	Topic 1	2	1	0	0	1	0
	Topic 2	0	0	1	0	0	0
	Topic 3	0	1	0	1	1	3

		Topic 1	Topic 2	Topic 3
$\mathbf{n}_2 =$	doc 2	1	0	3

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Collapsed Gibbs sampling

After one sweep through document 2:

w_1	boat	shore	bank		
z_1	1	1	1		
w_2	Zlatan	boat	shore	money	bank
z_2	3	1	3	3	1
w_3	money	bank	soccer	money	
z_3	3	3	2	3	

		boat	shore	soccer	Zlatan	bank	money
$n_v =$	Topic 1	2	1	0	0	2	0
	Topic 2	0	0	1	0	0	0
	Topic 3	0	1	0	1	1	3

		Topic 1	Topic 2	Topic 3
$n_2 =$	doc 2	2	0	3



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Other inference methods

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- Mean-Field Variational inference
(Blei, Ng and Jordan, 2003, Blei, 2013)



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- Mean-Field Variational inference (Blei, Ng and Jordan, 2003, Blei, 2013)
- Stochastic EM (Zaheer et al., 2015)

$$\hat{\theta}_d = \arg \max_{\theta \in \Theta} p(\theta, \phi | \mathbf{z}, \mathbf{w}) = \frac{n_{dk} + \alpha}{\sum^K (n_{dk} + \alpha)}$$

$$\hat{\phi}_k = \arg \max_{\phi \in \Phi} p(\theta, \phi | \mathbf{z}, \mathbf{w}) = \frac{n_{vk} + \beta}{\sum^V (n_{vk} + \beta)}$$

$$p(z_i = k | \mathbf{w}, \Theta, \Phi) \propto \hat{\theta}_{d(i),k} \cdot \hat{\phi}_{k,v(i)}$$



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Subsection 2

Model Evaluation



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- Parameter inspection (top words, relevance words, document topic distributions by other variables)
- Documents with high topic proportion: look at data



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- Parameter inspection (top words, relevance words, document topic distributions by other variables)
- Documents with high topic proportion: look at data
- Estimating held-out log likelihood (Wallach et al, 2009)



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- Parameter inspection (top words, relevance words, document topic distributions by other variables)
- Documents with high topic proportion: look at data
- Estimating held-out log likelihood (Wallach et al, 2009)
- Estimating topic coherence (Mimno et al, 2011)

$$C(V^{(t)}) = \sum_m^M \sum_l^{m-1} \log \frac{D(v_m^{(t)}, v_l^{(t)}) + 1}{D(v_l^{(t)})}$$

where $V^{(t)}$ is the set of the M top words $v_1, \dots, v_M$ and $D(v_m^{(t)}, v_l^{(t)})$ is the co-document frequency.



Top words are not enough

- Top words are useful for a first-pass label, but they are not evidence by themselves

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Top words are not enough

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- Top words are useful for a first-pass label, but they are not evidence by themselves
- Common words can hide the boundary between topics (e.g. stop words)



Top words are not enough

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- Rare or highly exclusive words can make a topic look cleaner than it is



Top words are not enough

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- Top words are useful for a first-pass label, but they are not evidence by themselves
- Common words can hide the boundary between topics (e.g. stop words)
- Rare or highly exclusive words can make a topic look cleaner than it is
- Compare:
 - Most probable words
 - Relevance words
 - Representative documents
 - Topic correlations and covariate patterns



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Representative documents

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- Read documents with high topic proportions



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- Read documents with high topic proportions
 - Check whether the topic label describes the whole document or only a few words



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 - Check whether the topic label describes the whole document or only a few words
 - Look for false positives: documents assigned to the topic for the wrong reason



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- Read documents with high topic proportions
 - Check whether the topic label describes the whole document or only a few words
 - Look for false positives: documents assigned to the topic for the wrong reason
 - Keep examples for paper writing:
 - Clear representative documents
 - Borderline cases
 - Document that reveals limitations



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Stability checks

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- Refit the model under alternative random seeds - label switching



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- Refit the model under alternative random seeds - label switching
 - Try nearby values of K



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- Refit the model under alternative random seeds - label switching
 - Try nearby values of K
 - Check sensitivity to preprocessing choices
 - Stop words, rare-word thresholds, stemming/lemmatization, named entities



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- Refit the model under alternative random seeds - label switching
 - Try nearby values of K
 - Check sensitivity to preprocessing choices
 - Stop words, rare-word thresholds, stemming/lemmatization, named entities
 - **Ask:** is the substantive conclusion stable?



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- Topic labels are interpretations, not estimated parameters



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- Topic labels are interpretations, not estimated parameters
 - Topic proportions are model-based estimates (with uncertainty)



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 - Topic proportions are model-based estimates (with uncertainty)
 - Explain instability, ambiguity, and failed validation checks when they matter



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- Topic labels are interpretations, not estimated parameters
 - Topic proportions are model-based estimates (with uncertainty)
 - Explain instability, ambiguity, and failed validation checks when they matter
 - Be explicit about what the model can and cannot support:
 - Exploratory description?
 - Measurement for downstream analysis?
 - Evidence for a historical or social-scientific claim?



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Model Practicalities



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- Structural Topic Models

- We can estimate the optimal K using log-likelihood.
Is this a good idea?



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- We can estimate the optimal K using log-likelihood.
Is this a good idea?
- **Alternative:** Think of K as the resolution of a map
- Evaluate when the relevant part of the model is good for the use case



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Section 3

Constraining topic models



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Why constraining?

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- Topic modeling is difficult using ordinary LDA



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 - The standard LDA is unsupervised - we might want to measure specific topics.



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- Topic modeling is difficult using ordinary LDA
 - The standard LDA is unsupervised - we might want to measure specific topics.
 - A simple approach is constraining topic models



Why constraining?

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- Topic modeling is difficult using ordinary LDA
 - The standard LDA is unsupervised - we might want to measure specific topics.
 - A simple approach is constraining topic models
 - Idea: Use the prior $p(\theta)$ to a priori define topics or documents



Example of constraining Φ

Constraining topic 2 to be a soccer topic:

		boat	shore	soccer	Zlatan	bank	money
$\Phi =$	Topic 1	0.35	0.35	0	0	0.15	0.05
	Topic 2	0.025	0.025	0.45	0.45	0.025	0.025
	Topic 3	0.025	0.025	0	0	0.45	0.45

Constraining topic 3 exists only in document 3:

		Topic 1	Topic 2	Topic 3
$\Theta =$	doc 1	0.96	0.04	0
	doc 2	0.6	0.4	0
	doc 3	0.05	0.35	0.6



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Pros and cons of constraining topic models

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- Pros

1. Reproducible
2. Transparent
3. Easier to diagnose problems
4. Can adapt the model to the research problem at hand



Pros and cons of constraining topic models

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- Pros

1. Reproducible
2. Transparent
3. Easier to diagnose problems
4. Can adapt the model to the research problem at hand

- Cons

1. Can break/work poorly in some settings
2. We get what we ask for



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The Swedish immigration discourse 1945-2019

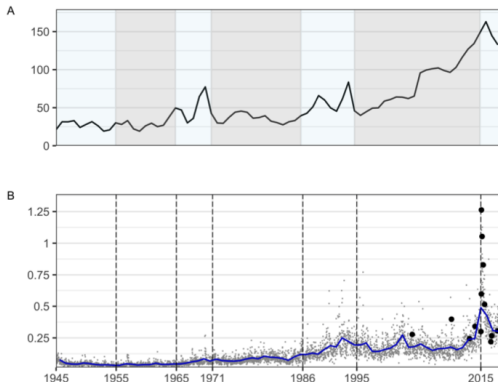


Figure: The number of immigrants to Sweden (A) and the saliency of immigration in the Swedish public discourse (B) (Hurtado Bodell et al., 2026, doi:10.1177/00491241241268453)



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Section 4

Structural Topic Models



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Why Structural Topic Models?

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- You have document-level covariates you want to include



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Why Structural Topic Models?

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- **Structural Topic Models**

- You have document-level covariates you want to include
- STM estimates covariate effects of the document-topic distribution (θ) by covariates X and topic-word distribution (ϕ) by covariates Y



Prevalence covariate vs content covariate

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- **Structural Topic Models**
- **Prevalence covariates**: explain how much a topic appears in documents
 - Example: year or party affects the prevalence of an immigration topic



Prevalence covariate vs content covariate

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- **Content covariates**: explain how a topic is expressed in words
 - Example: parties use different words when discussing the same policy topic



Prevalence covariate vs content covariate

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- **Prevalence covariates**: explain how much a topic appears in documents
 - Example: year or party affects the prevalence of an immigration topic
 - **Content covariates**: explain how a topic is expressed in words
 - Example: parties use different words when discussing the same policy topic
 - **Research question**: are groups talking about different things, or talking differently about the same thing?



- Modeling θ is done as

$$\theta_d \sim \text{LogisticNormal}(\mathbf{x}_d \gamma, \Sigma),$$

where $\mathbf{x}_d \in \mathbb{R}^P$, $\gamma \in \mathbb{R}^{P \times (K-1)}$, and $\Sigma \in \mathbb{R}^{(K-1) \times (K-1)}$

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- Interpretation of γ , the effect of a document covariate in using the topic.
- **Note!**
 1. Without covariates, it reduces to a correlated topic model



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- **Note!**
 1. Without covariates, it reduces to a correlated topic model
 2. Without covariates and $\Sigma = I$, it reduces to a standard topic model (ish)
 3. The parameters γ are interpreted to a reference category



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- Modeling ϕ is done as

$$\phi_{k,v,y} \propto \exp(m_v + \kappa_{k,v}^{(t)} + \kappa_{y_d,v}^{(c)} + \kappa_{y_d,k,v}^{(i)})$$

where $y \in \{1, \dots, A\}$, m_v is the marginal rate for word type v , $\kappa_{k,v}^{(t)} \in \mathbb{R}^{K \times V}$, $\kappa_{y_d,v}^{(c)} \in \mathbb{R}^{A \times V}$, and $\kappa_{y_d,k,v}^{(i)} \in \mathbb{R}^{A \times K \times V}$.



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- The effect of y_d on the word usage



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- The effect of y_d on the word usage
- Note!
 1. β is used instead of ϕ in the paper



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- The effect of y_d on the word usage
- Note!
 1. β is used instead of ϕ in the paper
 2. A Laplace (sparsity prior) is used to shrink κ parameters toward zero



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Example

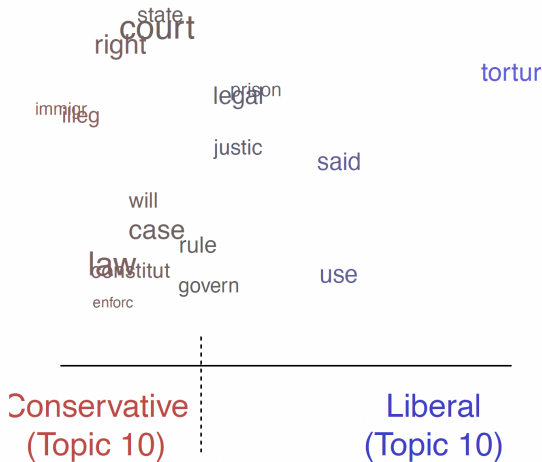


Figure: Difference between party word use (Roberts et al., 2019)



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Inference

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- STM (standard) is implemented using variational inference



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Inference

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- STM (standard) is implemented using variational inference
- Variational inference tends to **underestimate** the uncertainty of the parameters (Wang and Blei, 2018)



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Final takeaway

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Topic models are useful when they help you measure a textual pattern you can define, validate, and report transparently.

- The model proposes structure



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- The model proposes structure
- The researcher supplies interpretation



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Final takeaway

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- **Structural Topic Models**

Topic models are useful when they help you measure a textual pattern you can define, validate, and report transparently.

- The model proposes structure
- The researcher supplies interpretation
- The evidence comes from validation and careful reporting